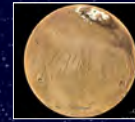


The Hubble telescope's 19th birthday was celebrated on the 24th of April 09



The Voyager spacecraft that was launched in 1977 is still travelling at 17 kilometers per second



Google Earth also has mapped the skies and terrain of the planet Mars

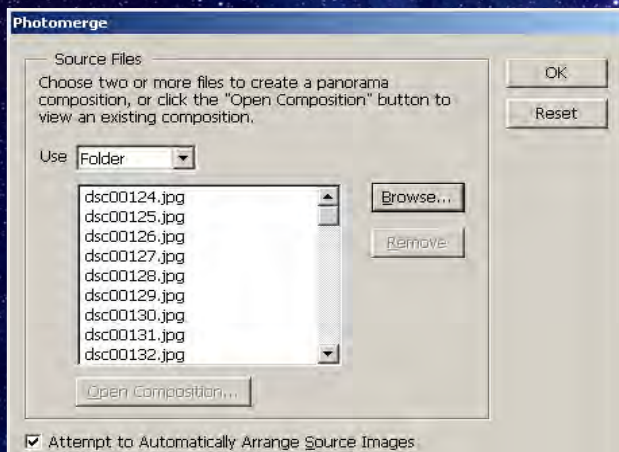
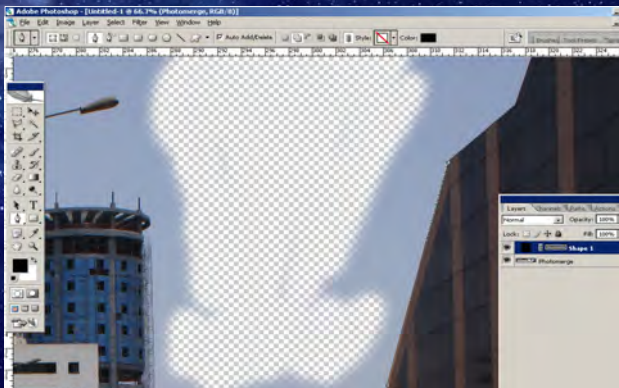


Figure 1: Adding the files and folders to the panorama

Complete the loop to make a shape. Then, right-click on the drawn path and click on Make Selection.

Once the image has been modified, it needs to be exported into the PNG image format. Click on File > Save As. Choose the format as PNG, give the file a name and save it. If you feel the output image is too large, you can also resize it down to something smaller. Remember the name of the file as you will require it to make the landscape.ini configuration file.



Clearing out the sky with the eraser and selection tool

Making the configuration file for the panorama

All the information that is to be used by Stellarium along with the panorama image is stored in a small text file called landscape.ini. The file is read by Stellarium when you select a landscape. This file is kept in the same folder as the panorama image. Use Notepad and create a file by the name of landscape.ini in the same folder as the panorama image.

A sample landscape file needs to contain the name for the landscape, the description, the image location itself and the altitude.

For example, we created a file called landscape.ini with these lines in it

```
[landscape]
name = Test panorama
author = Digit
description = Panorama test
type = spherical
maptex := panorama.png
```

This configuration should be fine if you shot all your photos starting from the east. If you haven't, then you have to rotate the landscape. First take a distinct point on the image that points out to the North (0 degrees). If required, draw a small mark on it on Photoshop. Looking at the entire panorama, judge how much to the left from the left of the image it is.



Figure 2: The rough mockup of the panorama in Photoshop

A simple formula is to be used : $270 + b - (360 * x / X)$. x is the distance in pixels between the left of the image to the spot you've chosen and X is to be the horizontal length of the entire image. Use Photoshop or any image editor to measure the distance in pixels. b is to be the heading you've chosen for the point. So if the spot, you've chosen is to be the North, then b will be 0.

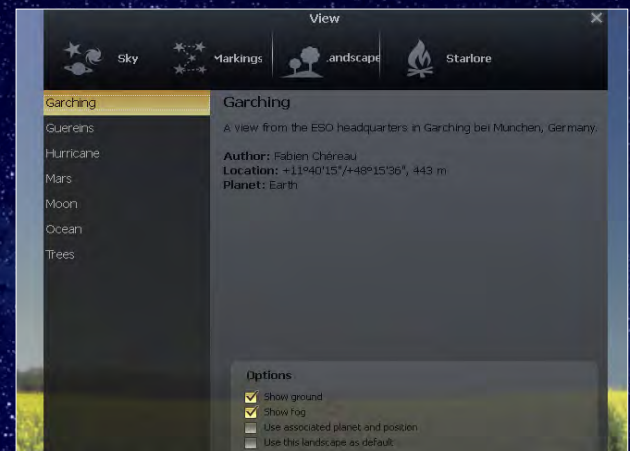
The result of this formula will be the degrees. Add the value to a line in the same text file

angle_rotatez = ____.

Inserting a panorama in Stellarium

Installing downloaded panoramas into Stellarium is simple. Unzip the zip file into the \Stellarium\Panoramas folder. Restart Stellarium and then use the vertical toolbar and click on the Sky and Viewing Options window. Click on Landscape and then choose the panorama you just installed.

In our case, we will simply copy the folder we created for the panorama into the Panoramas folder.



Choosing from a list of landscapes

Packing up the panorama

With the panorama ready to go, you can upload it to the Stellarium's Wiki. Create a thumbnail to put up on the page with the dimensions 200x114. Pack the files up into a ZIP folder with the folder, the image and the landscape.ini file in it. Also, it's a nice idea to include a README.TXT file which contains information about the landscape.

Once the panorama is ready, you can upload it to the Wiki located at <http://www.stellarium.org/wiki/index.php/Landscapes>. There are landscapes created by other users there for you to download and try out.

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